

GHULAM RASOOL MEMON

Civil Engineer

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SUMMARY

Civil engineering graduate with a focus on structural dynamics and the response of reinforced concrete structures to extreme loading, including blast and impact. Background in nonlinear finite element modelling.

ACADEMIC QUALIFICATION

- **BACHELOR OF ENGINEERING IN "CIVIL ENGINEERING" (B.E Civil)** 2021-2025
Mehran University of Engineering & Technology, Pakistan (First Division)

THESIS WRITTEN

- **Comparative Assessment of X-, K-, and V-Braced Reinforced Concrete Frames Subjected to Near-Field Blast Loading Using Nonlinear Finite Element Analysis.**

Final Year Project of Bachelor of Engineering in Civil Engineering at Mehran University of Engineering and Technology, Pakistan 2025. **(Thesis Supervisor: Engr. Dhanesh Kumar).**

- Developed and validated a nonlinear FE model of RC frames under near-field blast (350 kg TNT-equivalent, 5 m standoff), using the Concrete Damage Plasticity model and CONWEP blast loading.
- Validated against live-explosive RC column test data, with peak displacement matching within 5%.
- Compared bare, X-, K-, and V-braced configurations across seven monitoring nodes, using displacement, damage contours, and energy decomposition (strain, kinetic, viscous).
- Found bracing performance under blast to be highly location-dependent, diverging sharply from established seismic rankings
- Identified brace-to-frame connection geometry, not bracing type alone, as the governing factor in blast performance.

PROFESSIONAL EXPERIENCE

- **JUNIOR ENGINEER – TURKPAK INTERNATIONAL (PVT) LTD:** 20-April-2026 to Present
- Supervising execution of RCC structures (bridges, inlets, DRBs, and VRBs), ensuring work proceeds smoothly, safely, and per approved drawings and specifications.
 - Checking rebar placement, formwork, and concrete placement on-site, and coordinating with laboratory teams for concrete quality testing.
 - Coordinating with survey teams to verify levels and layout and identifying construction defects for timely corrective action to maintain quality and safety.
- **CIVIL ENGINEER (INTERN) – MOT MACDONALD PAKISTAN (PVT) LTD:** 01-Jun-2024 to 28-Jul-2024
- Assisted in on-site supervision of RCC works, following design and QA/QC standards.
 - Supported the smooth execution of civil, structural, architectural finishes, and MEP works as per IFC drawings, BOQ, and technical specifications.
 - Developed a 3D BIM (Revit) model of the mosque for design visualization and clash detection, as an internship assignment.
- **BIM/CAD MODELER (INTERN) – RCC CONSULTANTS & ARCHITECTS** 04-June-2022 to 24-July-2022
- Assisted in preparing tender and construction drawings for water supply, sewerage, and building works using AutoCAD.
 - Learned and CAD drafting for water, sewerage, and building projects, improving drawing accuracy and design documentation.
 - Improved drawing accuracy by updating and revising CAD files based on architect feedback.

RESEARCH INTERESTS

Structural Engineering | Nonlinear Finite Element Modelling | Blast and Impact Loading | Structural Dynamics | Concrete Technology | Sustainable & High-Performance Concrete | Corrosion and Concrete durability.

SOFTWARE SKILLS

ABAQUS (FEM) | ETABS | SAFE | Autodesk AutoCAD | Autodesk Revit | MS Office | Python

COMPLETED SEMESTER PROJECTS

- Determination of Stress and Strain Distribution Diagram of a RC Column with varying Transverse detailing Using ABAQUS.
- Linear Seismic analysis of 3-Story Residential Apartment on ETABS.
- Experimental Study of Sustainable Self Curing Mortar using Sugarcane Bagasse Ash as Cement Replacement and Sawdust as Internal Curing Agent.
- Complete 100% replacement of Coarse Aggregate with Burnt Brick chunks in Concrete.
- Evaluation of Construction Quantities and Costs for a RCC Commercial Building.
- Developed a comprehensive construction project schedule from drawings and BOQ, including WBS creation, activity sequencing, duration estimation, resource allocation, critical path analysis, and S-curve progress monitoring.
- Design of underground Water Tank by manual Calculations.

PUBLICATIONS/CONFERENCES

- **Memon, G.,** Kumar, D., Rind, T., Comparative Assessment of X-, K-, and V-Braced Reinforced Concrete Frames Subjected to Near-Field Blast Loading Using Nonlinear Finite Element Analysis. (**Under Review in Scientific Reports**).
- **Memon, G.,** Kumar, D., Hyder, Z., Ali, A., (2025). Evaluating the Structural Performance of Building Frame and Dual Frame System Incorporating Shear Walls. Aror University Research Journal, 2(2), 17–21. ISSN: 3106-9797.
- **Memon, G.,** Hussain, F., Kumar, D., (2025). Design And Simulation of Concrete Beams by Using Fly Ash as a Partial Replacement for Cement. Aror University Research Journal, 2(2), 01–05. ISSN: 3106-9797. <https://aurj.aror.edu.pk/aurj>
- **Memon, G.,** Kumar, D., Bhutto, S., Hussain, G., Examining the effect on Lightweight Concrete (LWC) by partially replacing Fine Aggregate with Crumb Rubber and Cement with Sugarcane Bagasse Ash (SCBA). 1ST INTERNATIONAL CONFERENCE ON SUSTAINABLE INNOVATIONS AND INFRASTRUCTURE DEVELOPMENT IN CIVIL ENGINEERING & TECHNOLOGY (ICSIID-2025), 59.
- Hussain R., Kumar, D., Ali, A., **Memon, G.,** Hassan A., Mechanical properties of Eco- Friendly Bricks utilizing Demolition waste and Fly ash. 1ST INTERNATIONAL CONFERENCE ON SUSTAINABLE INNOCATIONS AND INFRA STRUCTURE DEVELOPMENT IN CIVIL ENGINEERING & TECHNOLOGY (ICSIID-2025), 54.
- Hussain, F., Hyder, Z., Ali, A., **Memon, G.,** Complete Replacement of Coarse Aggregate with burnt bricks chunks in Concrete. 1ST INTERNATIONAL CONFERENCE ON SUSTAINABLE INNOCATIONS AND INFRA STRUCTURE DEVELOPMENT IN CIVIL ENGINEERING & TECHNOLOGY (ICSIID-2025), .63
- **Memon, G.,** Bhutto, S., Ali, S., Ali, A., Memon, Y., (2024). Comparative Structural Analysis of Building Frame and Dual Frame system by incorporating Shear Wall. 2nd International Multidisciplinary Conference on Emerging Trends in Engineering Technology-2024 (2nd IMCEET-2024), pp. 62–67, ISBN: 978-969-23814.

AFFILIATIONS/REGISTRATIONS

- Pakistan Engineering Council (PEC)
- Institute of Civil Engineers (ICE)

PROFESSIONAL TRAININGS, WORKSHOPS AND SEMINARS ATTENDED

- ABAQUS Finite Element Modelling Comprehensive Training.
- 2nd International Multidisciplinary Conference on Emerging Trends in Engineering Technology-2024 (2nd IMCEET-2024)
- 15th International Civil Engineering Conference (15th ICEC-2025).
- 2nd International Conference on Climate Change & Environmental Sustainability (2nd ICCCES-2025).
- 1st International Conference on Sustainable Innovations & Infrastructure Development (1st ICSIID-2025)
- Seminar on "Structural Integrity of Solid-State Recycling of Aluminum Chips".
- Seminar on "Utilization of Waste Materials in Concrete".

REFERENCES

- **ENGR. DHANESH KUMAR**
Lecturer/Supervisor - Mehran University of Engineering & Technology, Pakistan
Email: Dhanesh.kumar@muetkhp.edu.pk
- **DR. ABDUL QAYOOM MEMON**
Assistant Professor - Mehran University of Engineering & Technology, Pakistan
Email: qayoommemon@muetkhp.edu.pk
- **ENGR. TOUQEER ALI RIND**
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